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Investigating Mechanical and Acoustic Properties of 3D Printed Materials: A Focus on Printing Orientation

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ABSTRACT: Stereolithography (SLA) 3D printing technology is increasingly applied in geomechanics, petroleum engineering, geothermal, and CO2 storage research, particularly in fracture flow and mechanics. To understand the mechanical characteristics of 3D printed materials, we conducted laboratory experiments examining the effect of printing orientation on the properties of three specimens. These specimens were printed using a FormLabs (3B+) 3D printer at layering angles of 0°, 45°, and 90°, with a resolution of 100µm, and sized 1-inch in diameter and 2-inches in length. We performed triaxial compressive tests, ultrasonic elastic wave velocity measurements, and unconfined compressive strength (UCS) tests using an Autolab 1500 test frame. Incremental confining pressures from 1 MPa to 20 MPa were applied to explore static and dynamic mechanical properties. The UCS tests showed that all samples exhibited ductile behavior with similar stiffness and yield strength across different layering angles, demonstrating a stiffness of approximately 4 GPa and compressional wave velocities between 2555-2580 m/s at zero confining pressure. The findings confirm the mechanical and acoustic consistency of the samples, supporting their suitability for advanced geomechanical applications.

1. INTRODUCTION

Three-dimensional (3D) printing technology, also known as advanced manufacturing (AM), is a rapidly developing field that has revolutionized manufacturing processes across various industries including aerospace (Richter & Lipson, 2011), automotive (Rahim & Maidin, 2014), healthcare (Logan & Duddy, 1998), petroleum engineering (Li et al., 2021), and geomechanics (Phillips et al., 2021). This technology uses computer-aided design (CAD) to create three-dimensional objects by adding material layer by layer using almost any type of material such as polymers, and ceramics. Thermoplastic urethane and metals can also be employed as raw material (Gopinathan & Noh, 2018).

There are several techniques involved in 3D printing, including Fused Deposition Modeling (FDM), Stereolithography (SLA), Selective Laser Sintering (SLS), Digital Light Processing (DLP), Binder Jetting, and Material Jetting. Each technique has its advantages and disadvantages (Jandyal et al., 2022), and the choice of technique depends on the type of object being printed, the desired resolution, and the available materials. With the increasing popularity and availability of 3D printing technology, it is now relatively easy to create custom objects, prototypes, and even entire structures using 3D printing. Among the available techniques, SLA is known for producing high-resolution objects with smooth surfaces. SLA was first developed in the early 1980s by Kodama, 1981 and called Rapid Prototyping (RP).

One of the important mechanical properties of 3D printed objects is their anisotropy, which refers to the directional dependence of the physical properties of a material. In the case of SLA 3D printing, the objects produced can exhibit anisotropic behavior due to the specific printing process. The anisotropy in SLA-printed objects results from the layer-by-layer printing process, which creates a specific orientation of the cured resin layers. The orientation of these layers influences the mechanical properties of the printed objects, with some directions having higher strength, stiffness, or other mechanical properties than others. Furthermore, the degree of anisotropy in SLA-printed objects can also depend on specific printing parameters, such as the resin type, layer thickness, printing angle, and laser wattage. As a result, it is important to consider the anisotropic properties of SLA-printed objects when designing and using them for specific applications like geomechanics (Fractures) and petroleum engineering (porous media). This can involve optimizing the printing parameters and the orientation of the objects to achieve the desired mechanical properties.

In this study, the influence of printing layer orientation on the mechanical and acoustic properties were investigated by conducting confining and triaxial compressive tests along with acoustic compressional and shear wave velocity evaluations. We present experimental results demonstrating the orientations of the strength and stiffness of the cylindrical specimen under compression. Our results contribute to developing the proper use of 3D printing technology and provide valuable insights into the design and optimization of 3D printed components for geomechanical applications.

2. 3D-PRINTED SPECIMEN PREPARATION

The SLA printing process works by selectively curing photoreactive polymers with a laser. The process begins with a 3D CAD model with a Standard Tessellation Language (STL) file that is uploaded to the SLA printer. The printer then slices the model into thin layers and begins printing the bottom layer on a build platform in the X-Y plane by adding support structures that are automatically detected and created for models based on the overhang angle. The build platform is submerged in a liquid resin that is sensitive to UV light. A laser then scans the bottom layer, selectively curing the resin where the model requires material. The build platform is then lifted by one layer following the Z axis, and the process is repeated until the entire object is printed. Once the printing is complete, the object is removed from the built platform and rinsed in a reagent to remove any uncured resin. The object is then cured under UV light and heat to fully harden the resin and turn the part into a finished product. The resulting object has a high level of detail and accuracy, with smooth surfaces and precise dimensions.

In this work, we utilized the Formlab 3B+ 3D printer (Fig.1), which is well-suited for various professional settings including healthcare, engineering, and manufacturing.

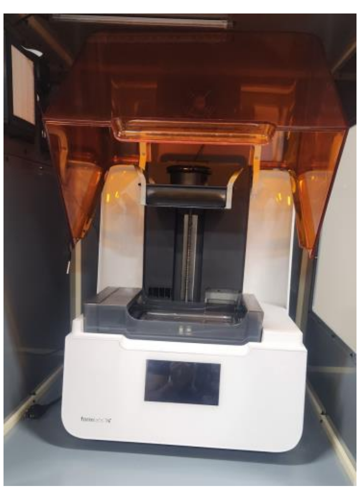

Fig. 1. Formlab 3B+ printer used in the study

V4 gray Polylactic Acid (PLA) resin was utilized to produce the printed samples. The SLA printer comprises a reservoir that holds a pool of liquid resin and a platform that gradually descends into the resin. As the platform reaches a specific depth, a laser beam is projected onto the liquid resin, creating the shape of the first layer of the object, as shown in fig.2. The laser triggers a chemical reaction to create monomer chains in the resin that solidify and fuse to the previous layer, forming a cohesive structure. After the first layer is complete, the build platform is then retracted slightly, and the process is repeated for the next layer. This layer-by-layer process continues until the entire cylindrical sample is printed.

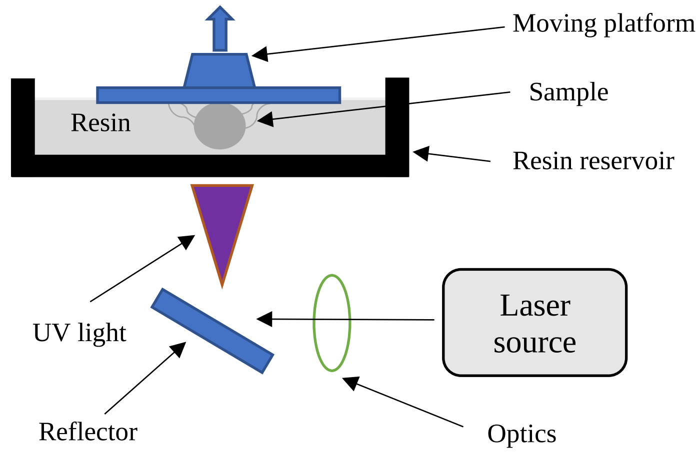

Fig. 2. Schematic of the SLA 3D printing process.

To obtain specimens with different transverse planes, the samples are printed at different angles (β), such as 0°, 45° and 90° as shown in Fig. 3. By placing the CAD cylinder at different angles during slicing, a range of transverse planes can be achieved, allowing for a more comprehensive understanding of the specimen's characteristics. The printing angle, (β), refers to the measurement of the angle between the principal axis of the specimen and the Y-axis of the printing platform.

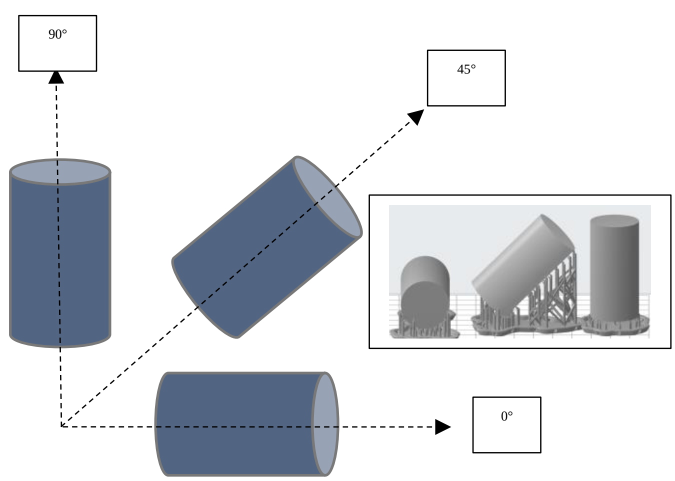

Fig. 3. Schematic view of the specimens placing orientation during the process of 3D printing

Once the printing process is complete, the specimen is removed from the printer, the support structure is removed (Fig. 4), and the specimen washed in a solvent (98 % isopropyl alcohol) to remove any excess resin.

Fig. 4. Support structures of 3D printing parts

The object is then cured under UV light and a temperature of 60 °C (Fig. 5) for one hour to fully harden the resin and make it strong and durable using Form_cure which is an automated post-curing system designed for SLA resin 3D printing enhances material properties, thereby optimizing the strength and performance of printed parts.

Fig. 5. Form_cure used to cure the samples

By adjusting the layer thickness, it is possible to achieve different levels of detail and precision in the printed object. For example, thicker layers can be used to print objects more quickly, but with less detail, while thinner layers can produce more intricate designs but take longer to print. Additionally, varying the layer thickness can affect the strength and durability of the printed object. In this study a 100µm layer thickness was chosen. The printing takes about 10 hours to print the three samples together.

Laboratory testing setup and procedures

In this study, we utilized the AutoLab-1500, located at the University of North Dakota and supplied by New England Research, Inc., for conducting sonic and mechanical experiments, as illustrated in Fig. 6. The AutoLab-1500 system is equipped to operate within two distinct pressure ranges, offering capabilities of up to 70 or 140 MPa, and can withstand temperatures up to 120 °C. The experimental setup allows for testing under diverse confining pressures, pore pressure, and temperature conditions. The system facilitates the implementation of conventional rock mechanics testing procedures, such as isotropic compression, uniaxial compression, triaxial compression, viscoelastic deformation with stress-controlled loading rate and strain-controlled loading rates, ultrasonic waveform measurements, including compressional and shear waves, and steady-state and pulse-decay permeability measurements.

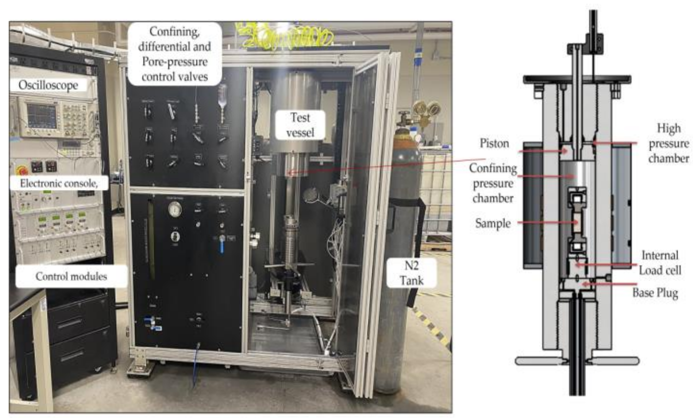

Fig 6: Autolab-1500 Equipment Overview: On the left, various components are displayed, while the right side showcases a schematic diagram detailing the pressure chamber, sample, and load cell assembly. Adapted from (Pothana et al., 2023)

2.1. *Experimental design*

We conducted three experiments: ultrasonic velocity measurements, triaxial testing, and unconfined uniaxial compression. The velocity and triaxial measurements were at various confining pressures, as outlined in the table 1. In the Triaxial testing, the axial load was incrementally increased up to 3 MPa for each confining pressure, and we recorded both axial and radial deformation during the loading and unloading phases. The detailed procedure for each experiment is explained in the following sections.

Table 1: Table shows the confining and differential pressures applied during the experiments.

Experiments	Confining Pressures	Differential Pressures
Ultrasonic Velocity Measurements	1, 2, 4, 6, 8, 10, 15 and 20 MPa	0 MPa
Triaxial testing	1, 2,4,6,8, 10, 15, and 20MPa	3 MPa at each confining pressure
Unconfined compressive strength	1 MPa	Until the sample reaches yield strength

2.2. *Ultrasonic velocity measurements*

In the sonic wave experiment, we tested core plugs by sending a compressional wave (P-wave) and two perpendicular shear waves (S-wave) along the vertical axis of the sample. We used ultrasonic end-caps with a native frequency of 750 MHz, which had a piezoelectric crystal stack generating the P-wave and two orthogonally polarized S-waves, namely S1 and S2 (the orientation of the S-waves polarization relative to the printing direction was not specifically considered in this study). The sample was attached to the source ultrasonic transducer velocity assembly using a shear wave coupler. The receiver ultrasonic transducer velocity assembly was connected to the core plug. This entire setup, including the specimen, was placed in a pressure vessel (Fig. 7). The AutoLab software was used to start the data acquisition process. Mineral oil was used in the pressure vessel to apply the confining pressure around the sample. During the experiment, we adjusted servo-controllers, filled the pressure chamber, advanced the axial piston, and configured the oscilloscope to obtain clear waveforms. Data were recorded at each of the various confining pressure values.

2.3. *Triaxial testing*

We conducted triaxial testing to assess axial deformation under different axial loads and confining pressures, as outlined in Table 1. We employed the same test setup (Fig. 7), used an ultrasonic wave velocity measurement apparatus, and measured axial and radial deformations using Linear Variable Differential Transducers (LVDTs), following ISRM guidelines (See Fig. 7). Before the experiments, calibration of axial deformations and stress measurements on known aluminum sample properties and a triaxial testing loading rate of 0.05 MPa/sec was performed. LVDTs were utilized to monitor axial and radial displacements (the direction of the radial sensors relative to the 3d printing direction was not considered in this study), calculating strains from specimen length and diameter changes. Two LVDTs monitored axial displacement, secured in a ring positioned about one radius above the upper end of the specimen (See Figure-7). Cores on extended rods attached to a second ring were mounted approximately one specimen radius below the lower end. The distance between these support rings was precisely measured prior to loading. As the specimen deformed, the separation between the rings changed, allowing for strain computation by dividing the displacement by the initial separation distance. The measured values were then employed to compute various static mechanical properties of the sample, including Young's modulus, bulk modulus, shear modulus, and Poisson's ratio.

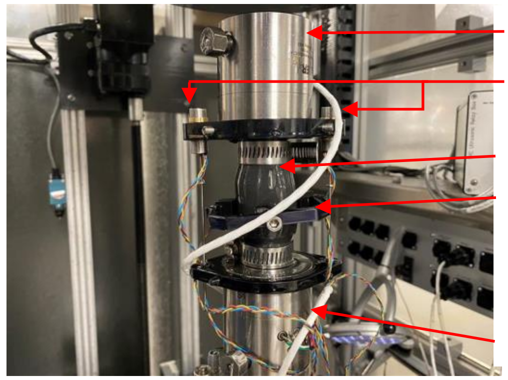

Fig. 7: Sample assembly

2.4. *Unconfined compression test*

Following ultrasonic and triaxial testing, the axial and confining pressures were reduced to a minimal level (1 MPa). Subsequently, axial loading was initiated with a loading rate of 0.006 mm/sec. Simultaneously, we recorded axial and radial deformations along with axial stress. A given test was concluded when the axial stress plateaued, indicating no further stress increments. Unlike real rock samples, the 3D-printed samples did not exhibit a stress drop after reaching ultimate yield strength. Instead, they demonstrated a stress increment reaching an asymptotic value. We defined this asymptotic limit as the ultimate strength of the sample.

3. RESULTS AND DISCUSSION

3.1. *Peak strength, Young's modulus, and Poisson's ratio*

After conducting Unconfined Compressive Strength (UCS) tests on the cylindrical 3D printed samples, noticeable bulging and buckling were observed in the samples (Fig. 8). The presence of bulging suggests lateral expansion of the samples under compressive loading, indicative of plastic deformation within the material. Additionally, buckling, characterized by sudden lateral deflection or bending, was evident in some samples, indicating a loss of structural integrity under the applied compressive force.

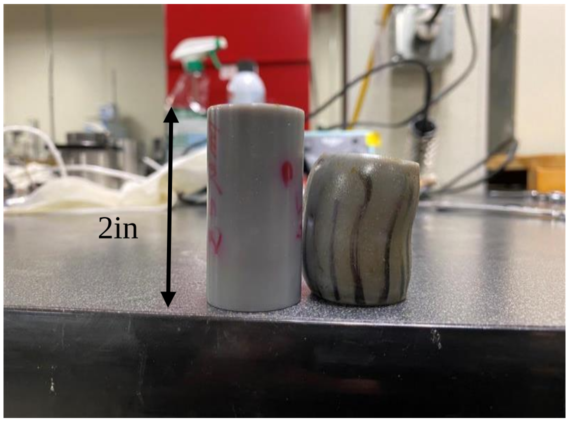

Fig. 8: Left: sample before testing. Right: sample after testing

The observed bulging led to Poisson's ratios exceeding 0.5 for all samples. This abnormal Poisson's ratio arises from outward expansion along the circumference of the samples under axial loading (Fig. 9).

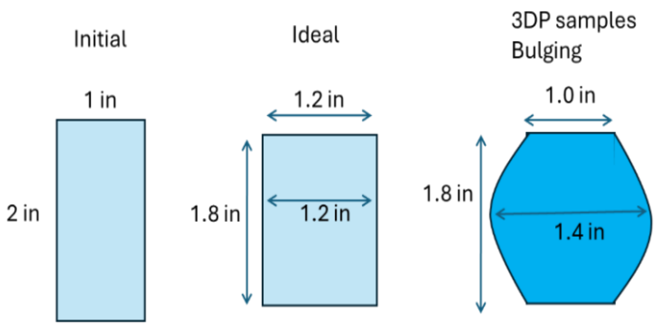

Fig. 9: Schematic Illustration of Barrel Deformation in Uniaxial Compression Strength (UCS) and Triaxial Testing. This figure depicts the non-uniform radial deformation observed along the length of the sample, emphasizing the qualitative nature of deformation rather than providing exact numerical values.

Regarding peak strength, the sample with zero-degree printing layers exhibited the highest value at 58 MPa, while those with printing layers at 45 degrees and 90 degrees displayed lower peak strengths of 54 MPa and 56 MPa, respectively. In terms of Young's Modulus, the zero-degree sample demonstrated the highest stiffness at 3.6 GPa, while the 45 and 90-degree samples exhibited a small difference 3.5 and 3.54 GPa respectively (Fig. 10).

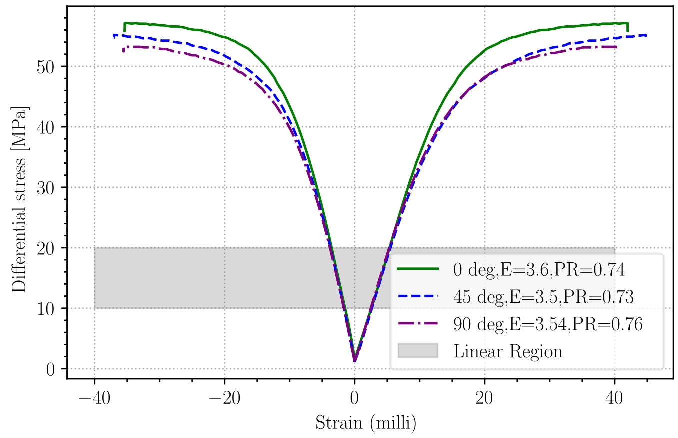

Fig. 10: Stress-strain profiles (left: radial, right: axial) for 0-, 45-, and 90-degree samples.

3.2. *Pseudo-Static Young's modulus and effective Poisson ratio*

Pseudo–static Young's modulus (PSYM), which refers to Young's modulus under confining pressure, showed significant variation with printing orientation. Samples printed at 0 degrees demonstrated higher PSYM values compared to those printed at 90 or 45 degrees, indicating increased stiffness perpendicular to the build direction.

This underscores the crucial role of printing orientation in determining the material's resistance to deformation under static loading conditions. (Fig. 11)

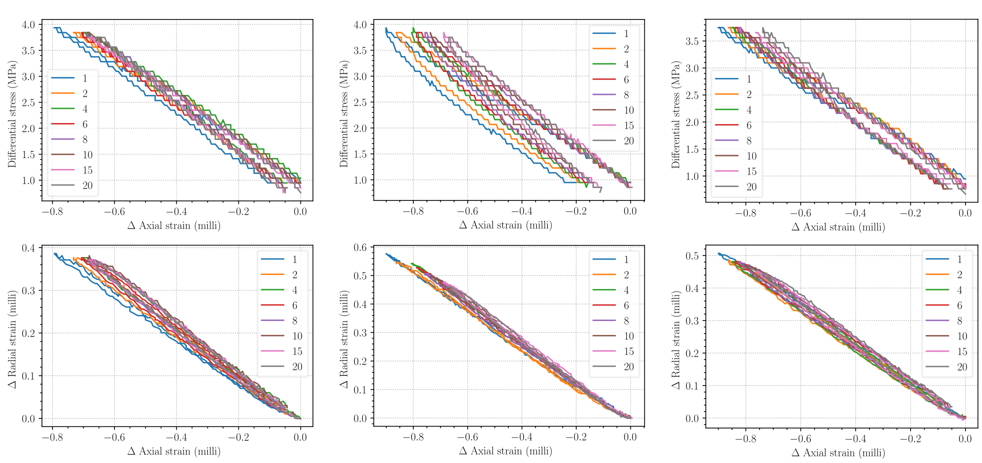

Fig. 11: Δ axial strain (upper figures) and Δ radial strain (lower figures) versus differential stress (upload and download) for 0, 45, and 90-degrees (left to right) samples respectively.

The observed differences in PSYM between printing orientations highlight the anisotropic nature of 3D printed materials. Samples printed at 0 degrees exhibited relatively higher stiffness along the printing axis, while those printed at 90 degrees demonstrated enhanced stiffness perpendicular to the printing direction. The lowest PSYM values observed for samples printed at 45 degrees further support the presence of anisotropic behavior in the material (Fig. 12)

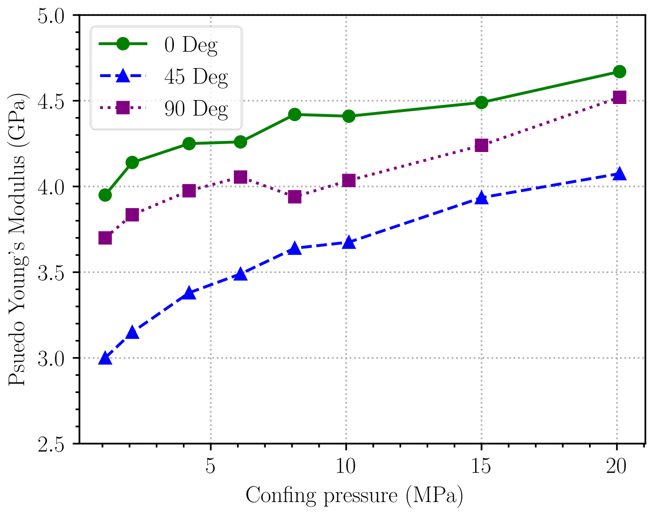

Fig. 12: Comparison of PSYM at different confining pressures for the 0, 45, and 90 degrees

Because of sample bulging, the effective Poisson's ratio values show abnormally high values that exceed 0.5 (Fig. 13).

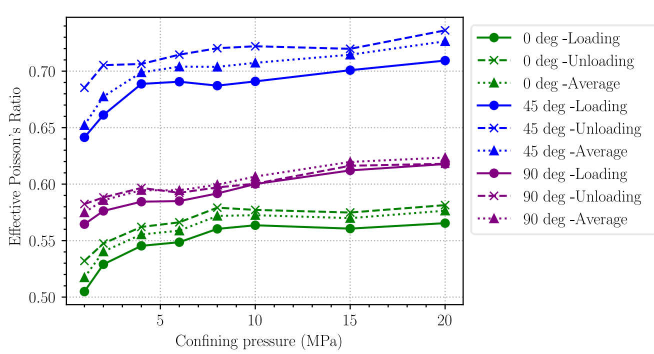

Fig. 13: Comparison of effective Poisson's ratio at different confining pressures for the 0, 45, and 90 degrees (loading and unloading)

3.3. *Acoustic results*

Overall, the trend observed across the three samples (as depicted in Fig. 14) indicates that the P-wave velocity tended to rise with increasing confining pressure. Notably, the 90-degree sample demonstrated the lowest P-wave velocity among the three. Moreover, the P-wave velocity of the 45-degree sample initially fell within an intermediate range until reaching 8 MPa, after which it surpassed that of the other samples, becoming the highest among the three.

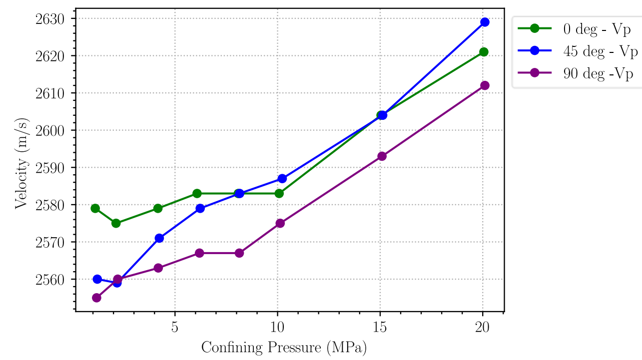

Fig. 14: P-wave velocity profiles for the 3 samples at different confining pressures

In general, the shear wave velocities Vs1 and Vs2 display a consistent increase across all samples with rising confining pressure, as illustrated in Fig. 15. Notably, the zero-degree sample consistently exhibited the highest values for shear waves. However, the S2-wave velocity of the 90-degree sample started as the lowest until reaching 10 MPa, after which it exceeded that of the 45-degree sample and eventually surpassed it. Conversely, for the S1 velocity, the 90-degree sample initially recorded the lowest value but surpassed the 45-degree sample after 10 MPa.

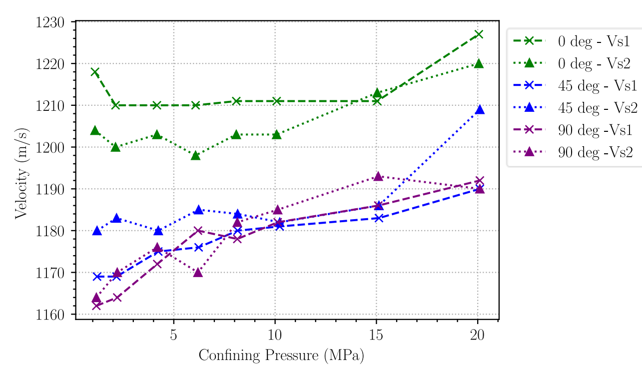

Fig. 15: S1 and S2-wave velocities profiles for the 3 samples at different confining pressures

The dynamic Poisson's ratio for the three samples remains relatively stable despite rising with confining pressure. Among these, the sample at 0 degrees exhibited the lowest Poisson's ratio, recorded at 0.36, while the ratios of the remaining samples were slightly higher at 0.3655 (Fig. 16).

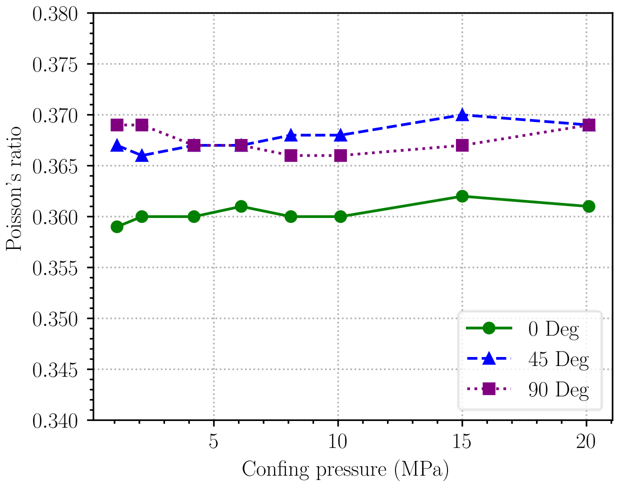

Fig. 16: Dynamic Poisson's ratio for the 3 samples at different confining pressures

4. CONCLUSION

Based on the comprehensive examination of Unconfined Compressive Strength (UCS), Young's modulus, and acoustic properties of 3D printed samples with varying printing orientations, several key conclusions can be drawn.

Firstly, the UCS tests revealed noticeable bulging and buckling in the samples, indicating plastic deformation and loss of structural integrity under compressive loading. The UCS and Young's modulus values do not vary significantly with printing orientation, with samples printed at zero degrees demonstrating the highest UCS and stiffness perpendicular to the build direction. This shows that printing orientation is irrelevant in determining mechanical properties.

Furthermore, the analysis of Pseudo-static Young's modulus (PSYM) emphasized the influence of printing orientation on the material's resistance to deformation under static loading conditions. Samples printed at zero degrees exhibited increased stiffness along the printing axis, while those printed at 90 degrees demonstrated enhanced stiffness perpendicular to the printing direction, with samples at 45 degrees exhibiting the lowest PSYM values.

In terms of acoustic properties, the trends observed in P-wave velocity and shear wave velocities (Vs1 and Vs2) with escalating confining pressure provided insights into the material's behavior under different loading conditions. The variations in velocity across samples further underscored the significance of printing orientation in determining acoustic properties.

Overall, this study highlights the importance of considering printing orientation in optimizing the mechanical and acoustic performance of 3D printed materials. Future research could delve deeper into understanding the underlying mechanisms driving the observed variations and explore strategies to mitigate the anisotropic behavior for enhanced material performance across different applications like fracture mechanics and coupled geomechanics and fluid flow.

These observations point to potential limitations in the structural performance of the 3D printed material under compressive loading conditions, highlighting areas for further investigation and optimization in order to enhance the overall mechanical properties and performance of the printed components. Furthermore, this study opens up avenues for potential applications of 3D printed samples in geomechanics, particularly in understanding fracture mechanics and behavior under various loading conditions. The suggestion of utilizing 3D printing technology to create intersected rough fractures within samples presents an intriguing opportunity for advancing research in this field.

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